Project Scheduling

In project management, a schedule is a listing of project milestones, activities and deliverables usually with intended start and finish dates and estimates in terms of resource allocation, budget and duration.

Project Schedules: Before work commences on a project, or on a stage of a larger project the project plan must be developed showing dates when each activity should start and finish and when and how much of each resource will be required. This refined plan is called project schedule. Creating a project schedule includes four steps.

Project Scheduling contd..

The first step is to decide what activities need to be carried out and in what order they are to be done. This is known as ideal activity plan development.

The second step will be activity risk analysis or identifying potential problems. The ideal activity plan may be altered due to the risk.

The third step is resource allocation. The availability of resources might place constraints on when certain activities can be carried out and the ideal plan might be changed.

The final and the fourth step is schedule production. The final project schedule is documented with resources allocated to each activity with start and finish dates.

Projects and Activities

Defining Activities: A project is composed of a number of interrelated activities.

- A project may start when at least one of its activities is ready to start.
- A project will be completed when all of the activities have been completed.

An activity must have a clearly defined beginning and a clearly defined end point and produces a deliverable.

If an activity requires a resource that resource requirement must be known in advance and assumed that resource is needed throughout the duration of the activity.

Projects and Activities contd..

- The duration of an activity must be forecastable.
- Some activities might require that others are completed before they can begin.

Identifying activities: There are three approaches to identify the activities or tasks in a project.

(i) Activity based approach (ii) Product based approach and (iii) Hybrid approach

The activity based approach

Create a list of all the activities that the project involves. This can be performed by involving whole project team

It will be helpful to subdivide the project into the main life cycle stages and consider each stage separately

Another way of creating a task list is to create a work breakdown structure (WBS). In WBS high-level tasks are identified first and then breaking down each of these high-level tasks into a set of lower-level tasks is performed

Once the activities have been identified they need to be sequenced in such a way that tasks that need to be completed before others start must, be scheduled first

The product based approach

It consists of producing a product breakdown structure and product flow diagram (PFD)

The PFD indicates for each product, which other products are required as inputs.

The PFD can be transformed into an ordered list of activities by identifying the transformation that turns some products into others.

This approach is appropriate if using Structured Analysis and Design Methodology (SADM) or USDP (Unified Software Development Process) which clearly specifies for each step or task, each of the products required or activities required to produce it.

The product based approach contd..

In USDP, products are referred to as artifacts and the sequence of activities needed to create them is called a workflow.

USDP emphasizes that the processes are iterative; hence it may not be possible to map an USDP process directly onto a single activity in a network

All projects will need to have some fixed milestones or time-boxes if project is to be completed by a planned delivery date

The larger activities with fixed delivery dates would be the basis of the activity network

The hybrid approach

It uses both WBS and product based approach

In this WBS may be based on the project's products instead of based entirely on structuring of activities.

The products in turn are based on a simple list of final deliverables and for each deliverable, a set of activities required to produce that product

Thus in the hybrid approach of identifying activities in a project, additional levels are introduced by structuring both products and activities

A framework dictating the number of levels and the nature of each level in the structure may be imposed on WBS

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The hybrid approach contd..

For example in the project methodology developed by IBM called (MIIP (managing the Implementation of the Total project), IBM recommended that the following five levels should be used in WBS

Level 1 : Project

Level 2: Deliverables as software, manuals and training courses

Level 3: Components which are the key work items needed to produce deliverables such as the modules and tests required to produce the system software.

Level 4: Work-packages or collections of related tasks, required to produce a component.

Level 5: Tasks, which are performed with the responsibility of a single person.

Project Time Management (PTM)

PTM includes the processes required to accomplish timely completion of the project. The PTM processes include the following:

1. Activity Definition: Identifying activities that need to be performed to produce deliverables.
2. Activity Sequencing: Identifying and documenting dependencies among scheduled activities.

Project Time Management (PTM) contd..

3. Activity Resource Estimation: Estimating the type and quantity of resources required to perform each scheduled activity.
4. Activity Duration Estimation: Estimating the number of work periods required to complete each scheduled activity.
5. Schedule Development: Create project schedule after analyzing activity sequences, durations and resource requirements.
6. Schedule Control: Controlling changes to the project schedule.

Project Time Management (PTM) contd..

These processes interact and overlap. Each process needs effort from one or more persons or groups.

For projects with smaller scope, these processes are tightly linked and can be performed by only a single person over a short period of time

The develop project management plan process plans the work involved in these processes.

It also produces a schedule management plan that provides criteria to develop and control the project schedule.

Overview of Project Time Management

